

Network Traffic Analysis Using Wireshark

Objective

To capture and analyze live network traffic using Wireshark and understand DNS, TCP, and TLS protocols.

Tools Used

Wireshark 4.6.6, Windows 11, Wi-Fi Network

Protocols Analyzed

DNS, TCP, TLS/HTTPS, QUIC

Findings

- Captured DNS queries and responses to observe domain name resolution.
- Identified TCP SYN packets and analyzed the TCP three-way handshake.
- Examined TLS Client Hello and Server Hello packets.
- Observed encrypted HTTPS communication and SNI information.
- Performed packet filtering and protocol analysis using Wireshark.

Conclusion

Successfully captured and analyzed live network traffic using Wireshark. Gained practical experience in DNS analysis, TCP connection establishment, TLS encryption, and cybersecurity fundamentals. This project demonstrates hands-on packet analysis and network troubleshooting skills.